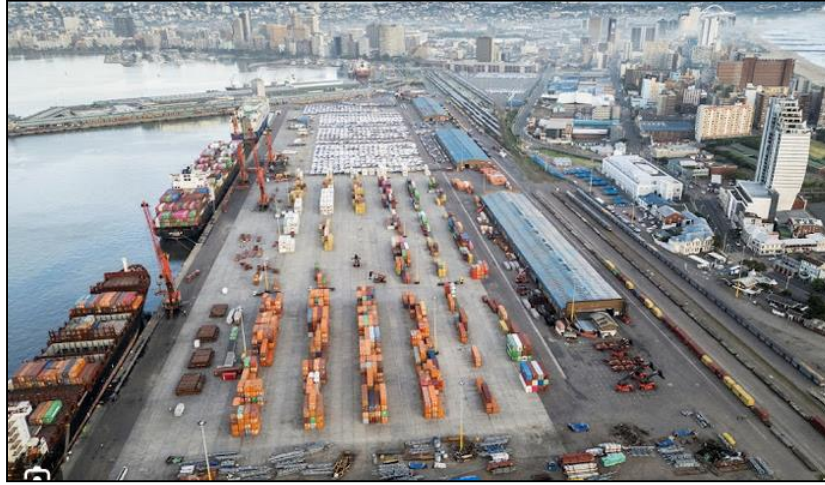




RAIL NETWORK STATEMENT RESPONSE

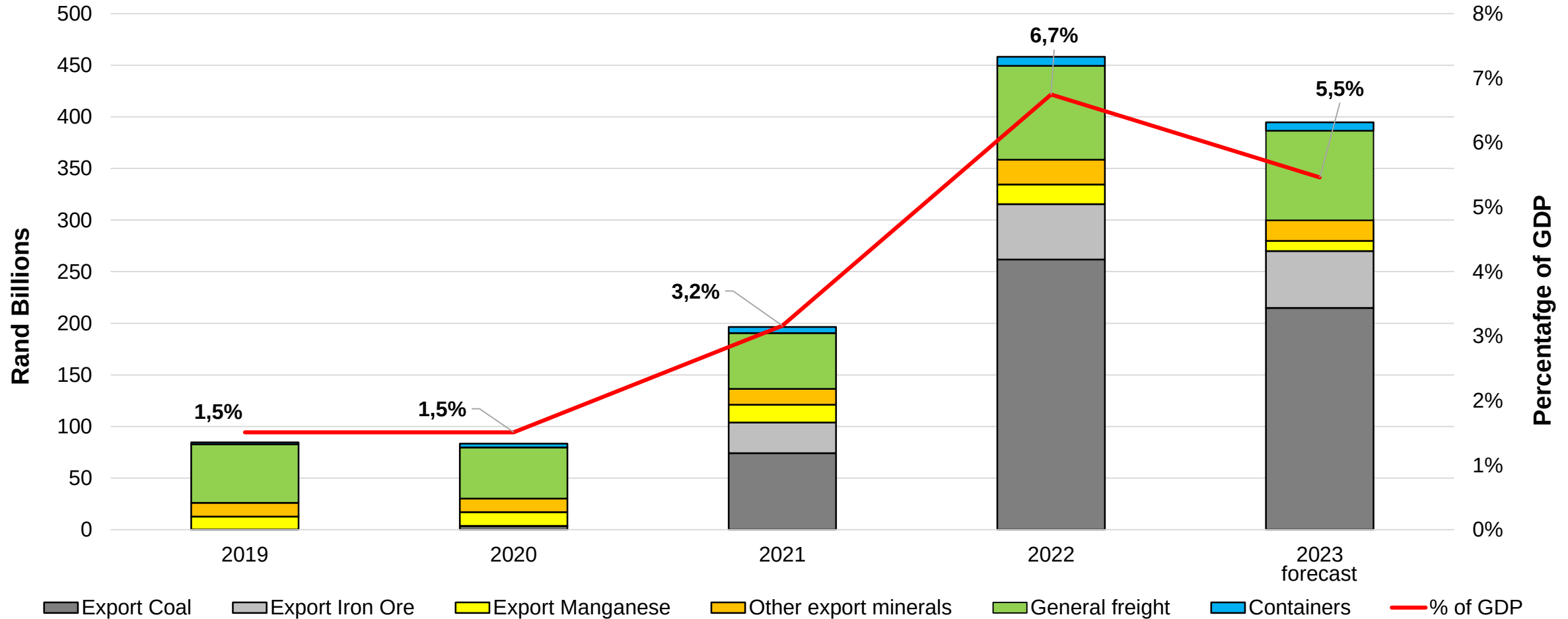
Prof Jan Havenga
Dr Zane Simpson
Henk Neethling

May 2024

Network statement response

- The problem
- The promise
- The emergence of a rail to road strategy
- Why are we here? (Regulation history)
- The prize
- What now

Transnet is damaging the economy – Rail disruptions and failures



There is a link between the Billion Rand per day loss and the R30 million per day interest overcharge (which can be greatly reduced and is a general waste).⁵

The end goal is to mobilise rail correctly

Figure 7: Market share of rail %

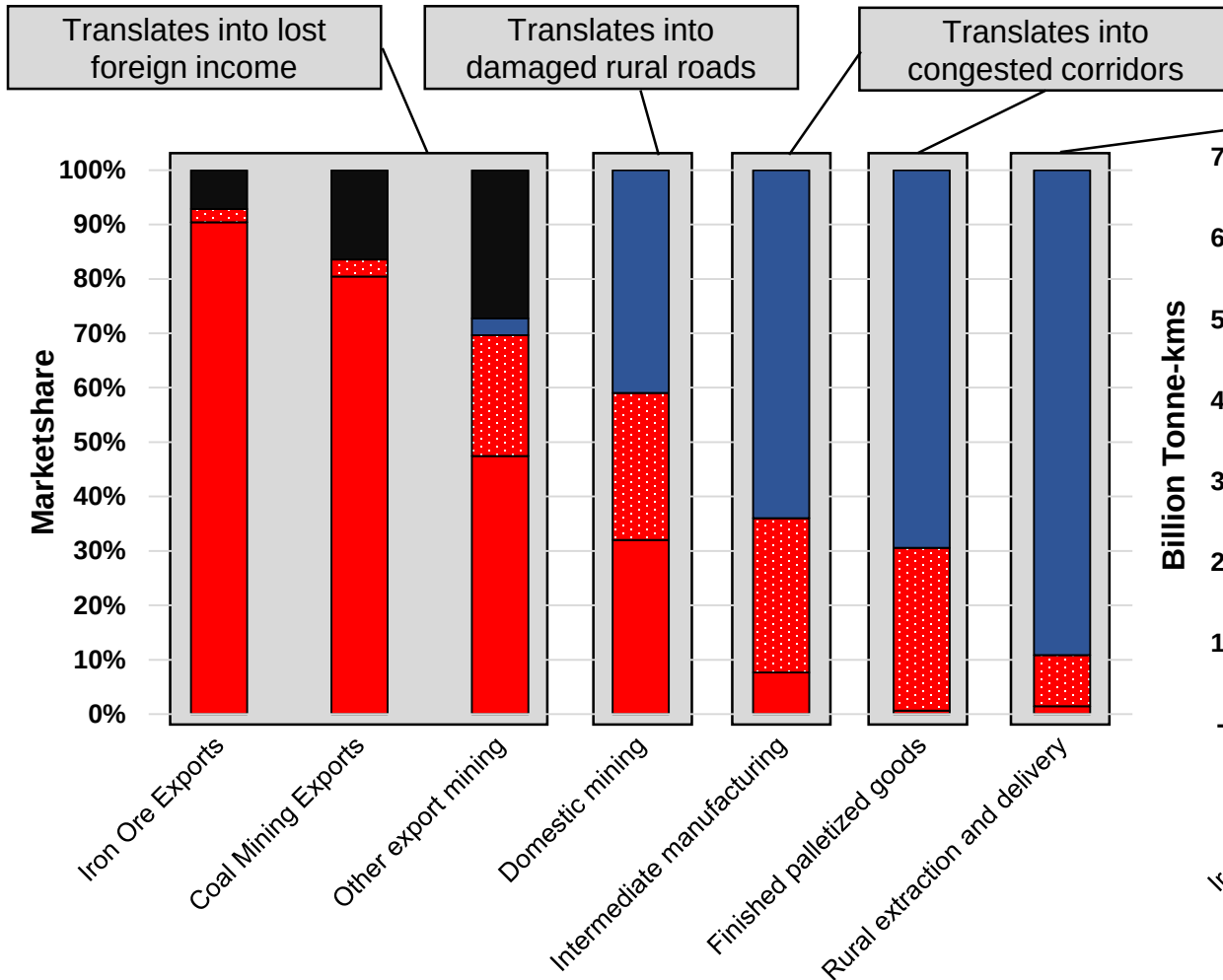
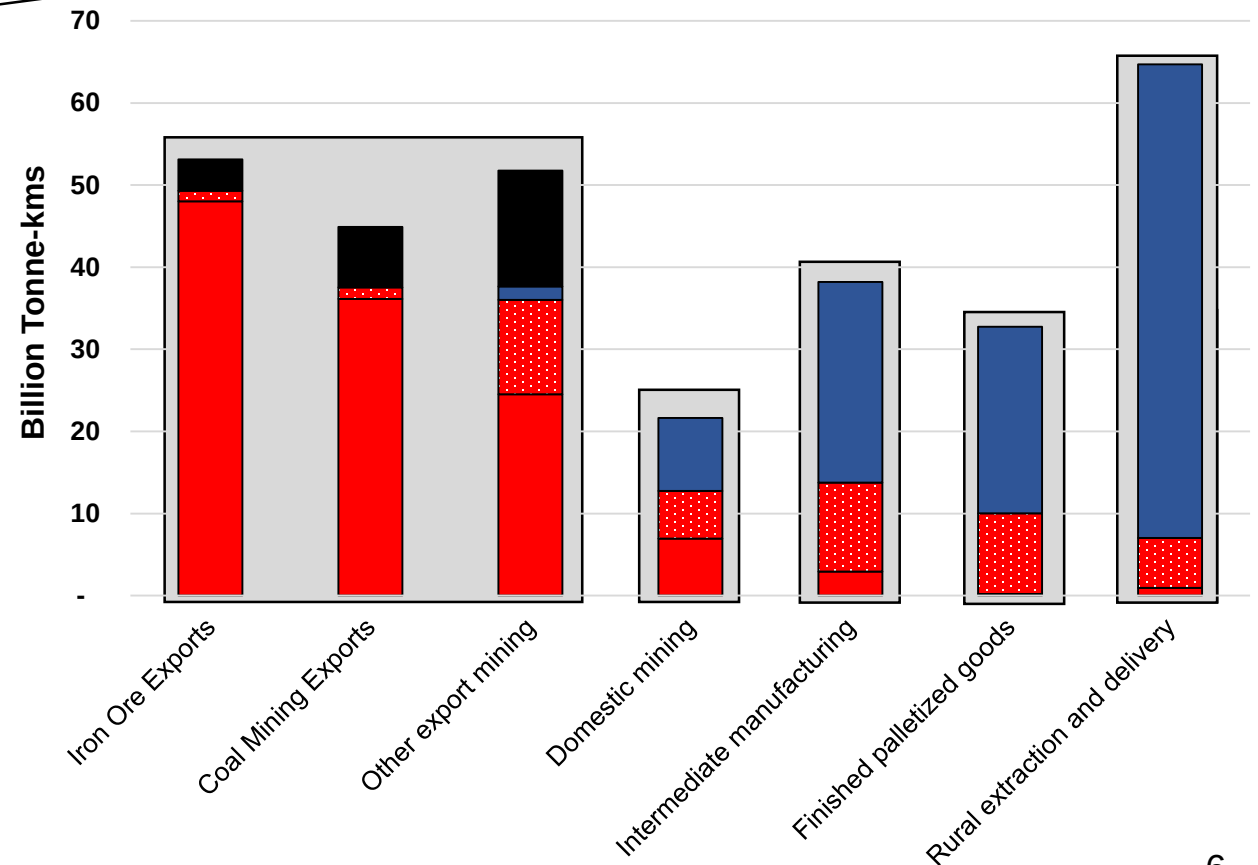


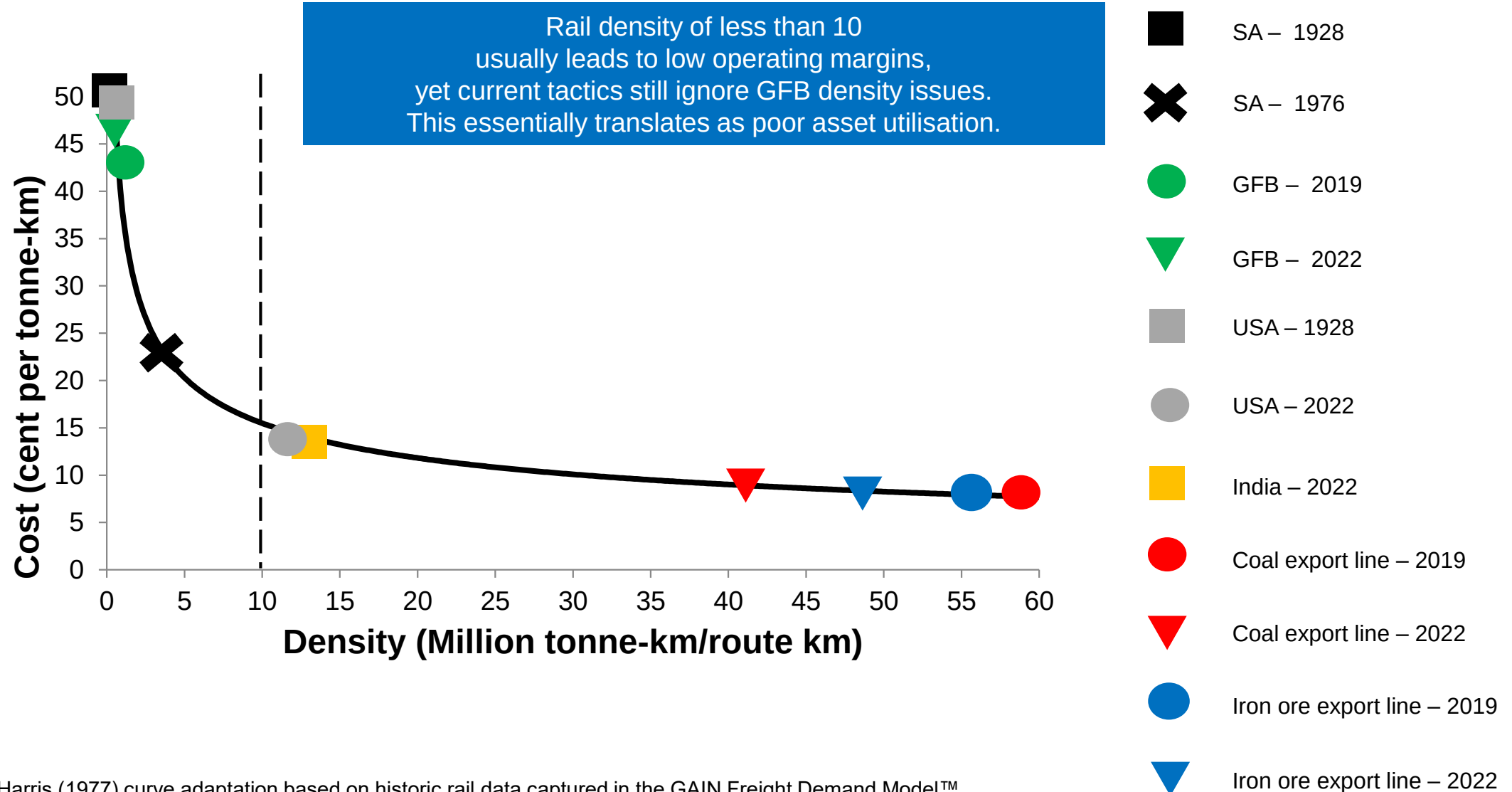
Figure 8: Market share of rail tonne-kilometres



Current rail Road freight that should be on rail Road freight that should be on road Missing lost volumes

Source: GAIN Freight Demand Model™

GFB density is a disaster

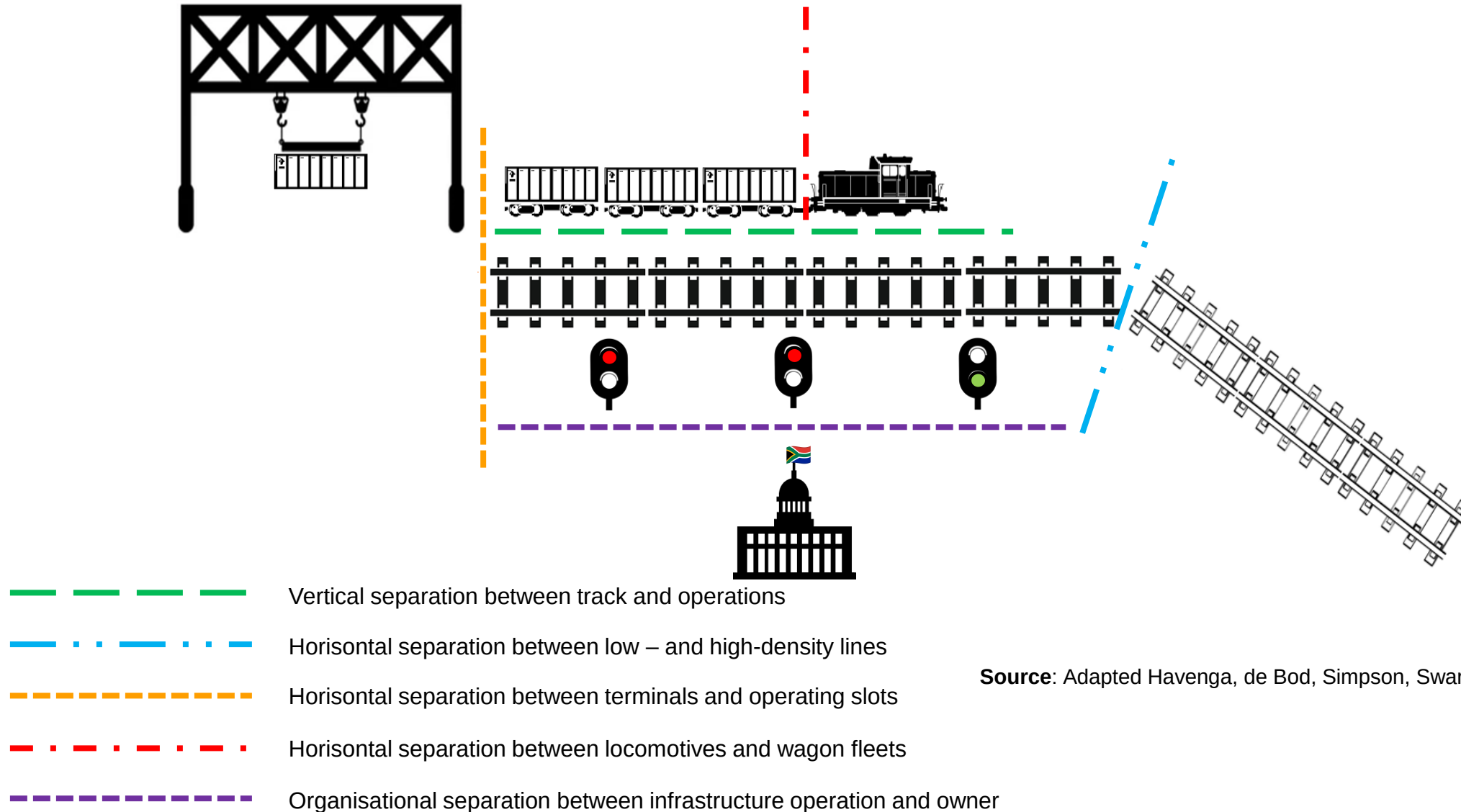


Source: Harris (1977) curve adaptation based on historic rail data captured in the GAIN Freight Demand Model™

Network statement response

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What emerged is the promise of separation

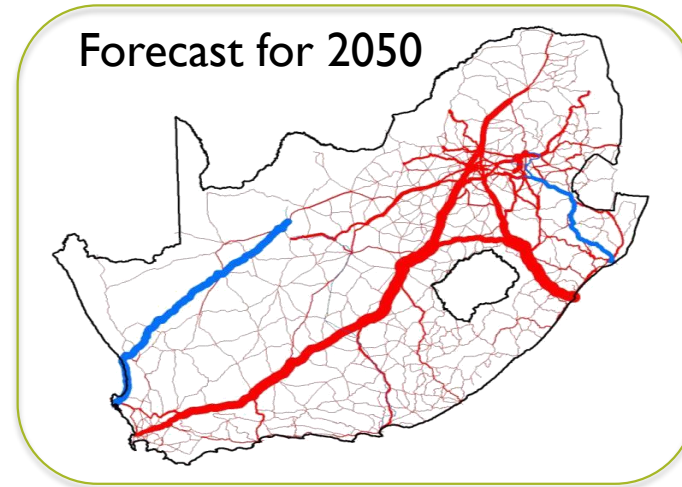
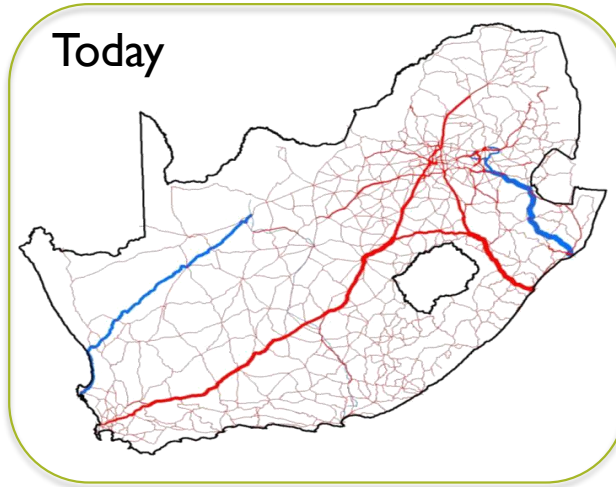


Source: Adapted Havenga, de Bod, Simpson, Swarts & Witthöft (2021:33)

Network statement response

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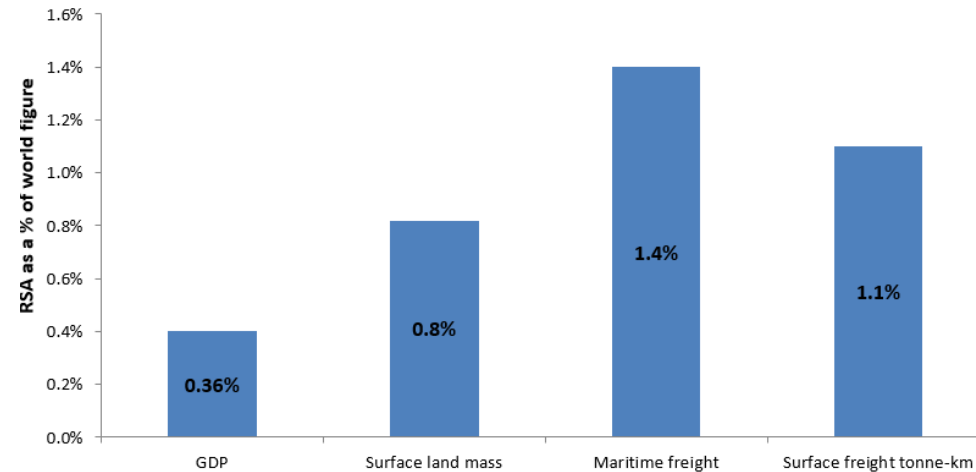
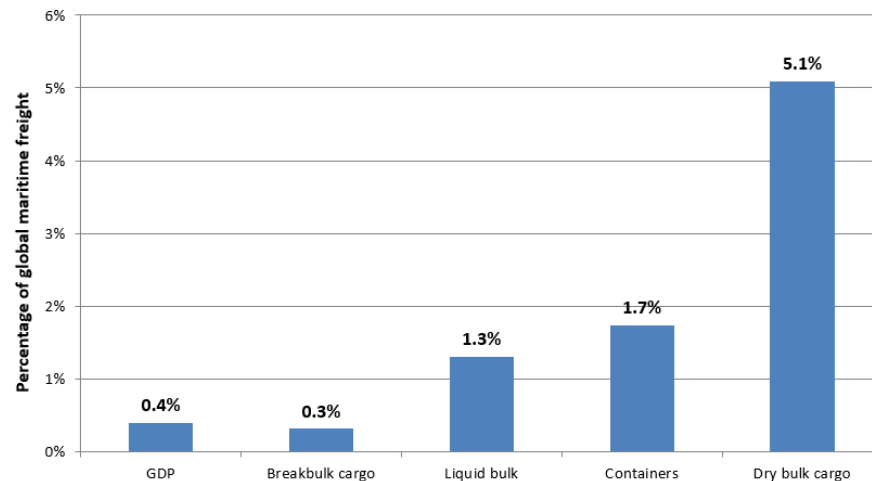
South Africa's freight demand is high



GDP

- South Africa
\$0.35 trillion
- France/Germany
\$6.26 trillion

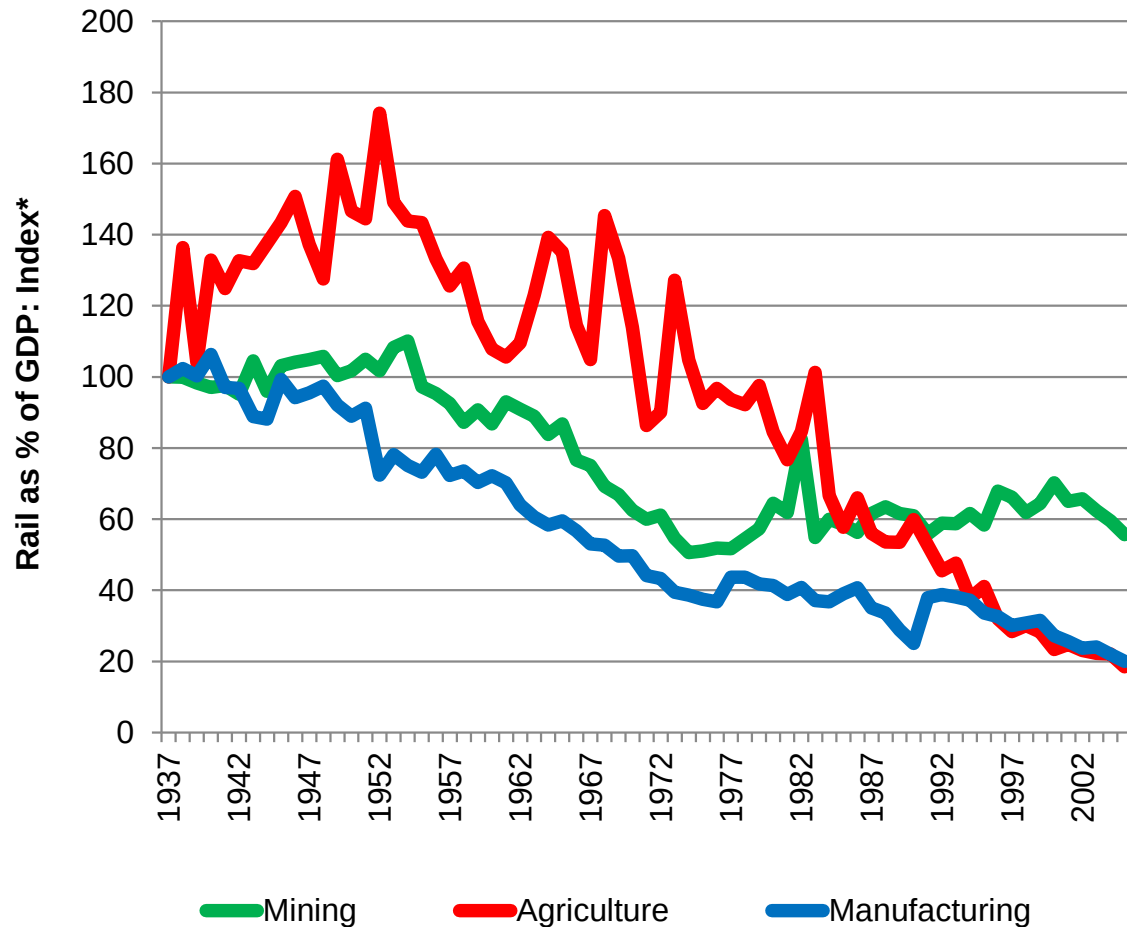
Note: In the maps above, the red lines indicate all freight flows and the blue lines indicate dedicated export line flows.



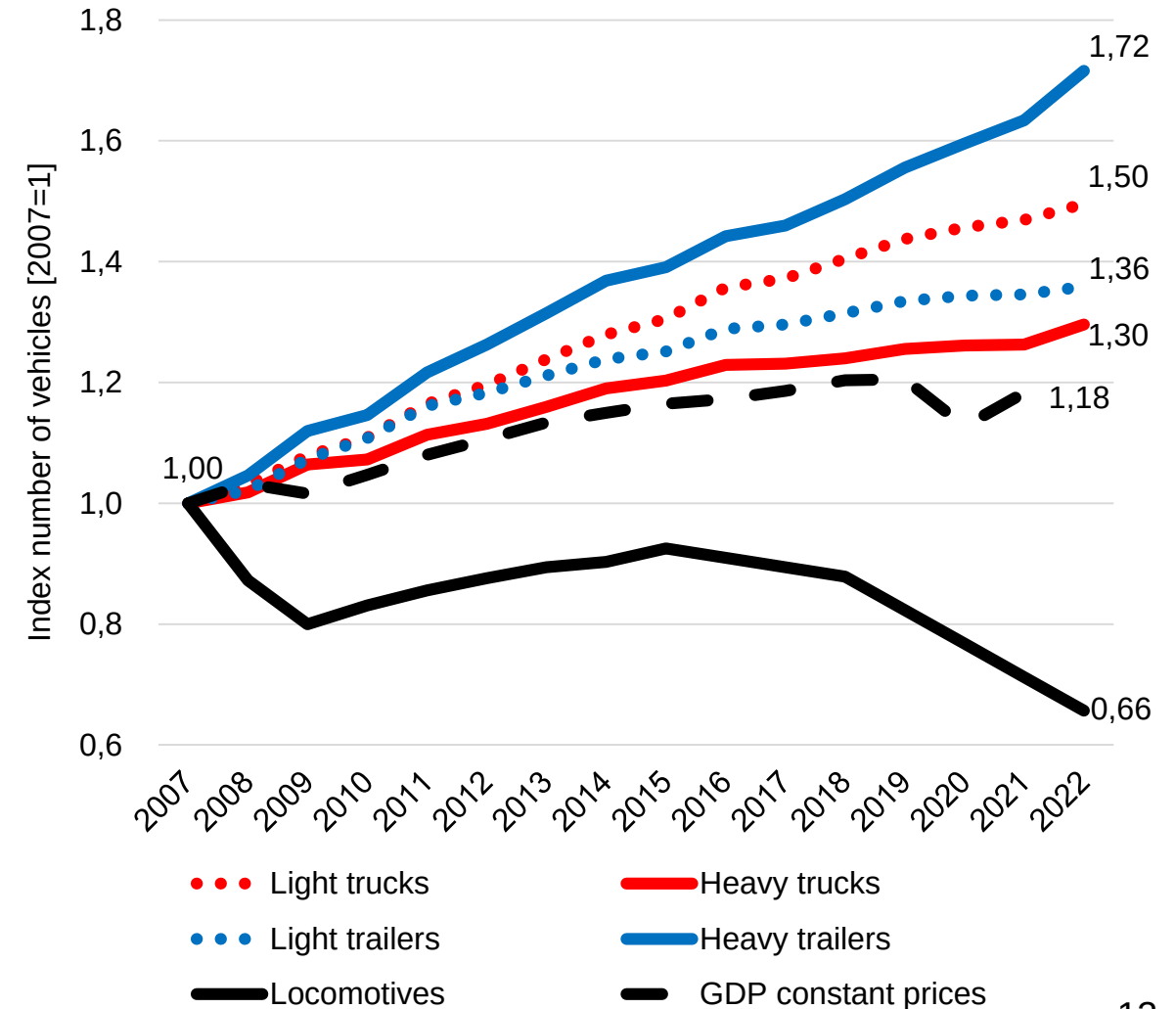
SA's landmass is about equal to Germany and France combined, while its GDP is nearly 18 times smaller

Source: GAIN Freight Demand Model™

With a slow and sometimes unnecessary move to road supply

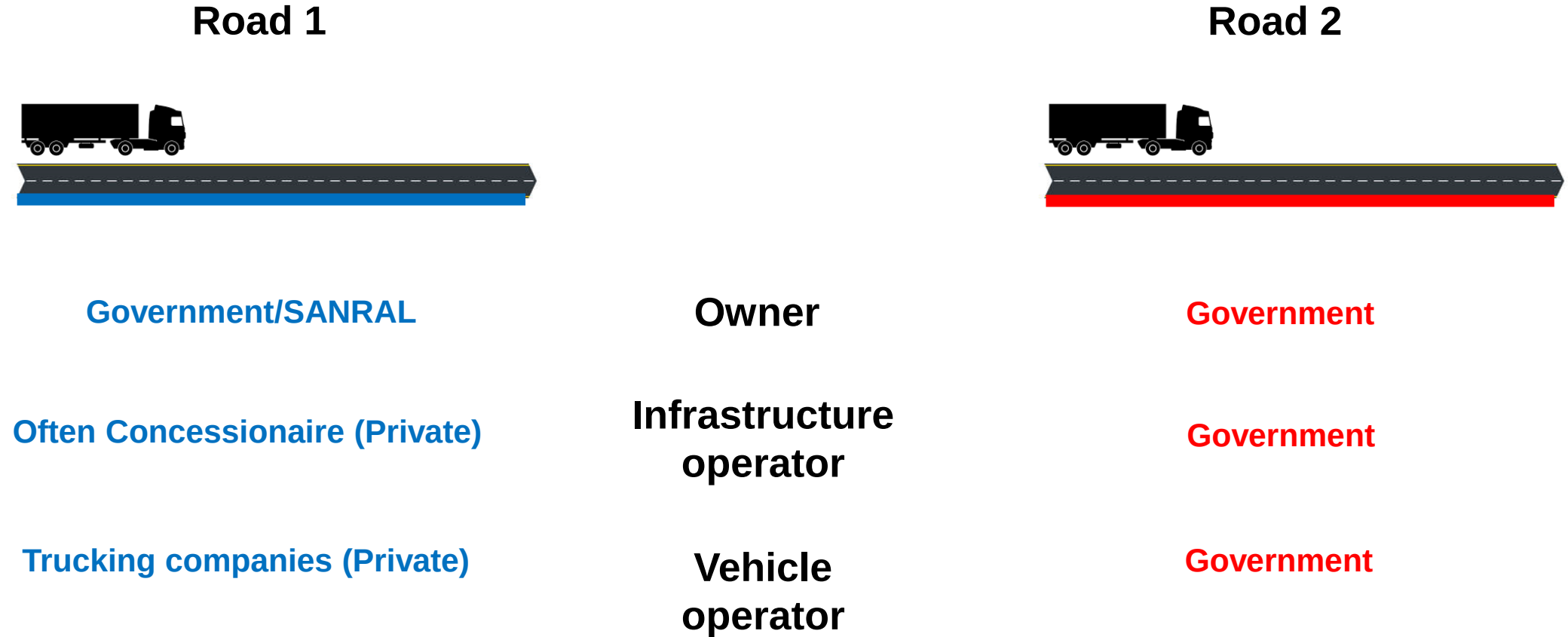


*Indexed correlation of the relationship between rail transport and physical production in the



Source: GAIN Freight Demand Model™; Havenga et al. (2021) with 2022 data for eNaTIS vehicles, Transnet locomotives and World Bank GDP

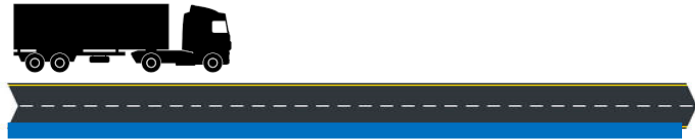
Let us imagine



Road 2, however, has many social and economic benefits.

And we're, of course, talking about our railway

Road 1



Government/SANRAL

Often Concessionaire (Private)

Trucking companies (Private)

Owner

Infrastructure
operator

Vehicle
operator

Rail network



Transnet

Transnet

Transnet

Access charges

Road 1



- **Current Access Charge**

- Fuel levy
- License fees
- Toll fees



- No return on assets (ROA)
- Some cross subsidy questions

Rail network



- **Proposed Access Charge**

- ROA
- Maintenance
- Profit



- Asset is extensively leveraged and heavily overvalued
- Not subsidised
- Rampant internal cross-subsidisation

Whether the fuel levy is an access charge or just a “random tax” is a point of debate.

Cape Town to Johannesburg example

Current road

- **Fuel levy:** R 3.84 per litre x 1 500 litres
~ R 5 800
- **License fees:** ~ R 100
- **Toll fees:** ~ R 800
- **Cost:** R 5 800 + R 100 + R 800

~ R 7 000

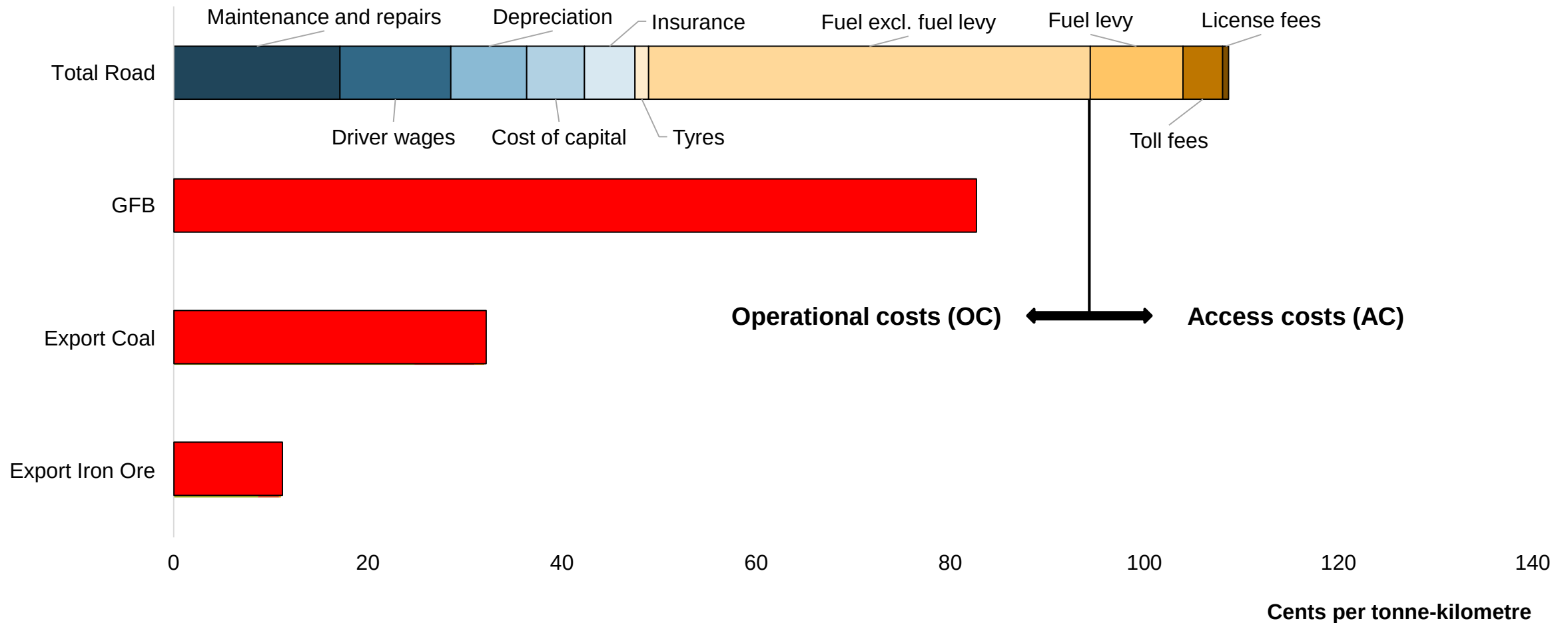
With 20c access charge

- **I.e. replace fuel levy, license fees and toll fees with 20 cents per tonne-km access charge**
- **Tonne-kms:** 57 tonne interlink x 1 400 km
~ 80 000 tonne-kms
- **Cost:** 80 000 tonne-kms x R 0.2 per tonne-km

~ R 16 000

This obviously too simplistic, due to the “work” of the infrastructure.

Road and rail cost components - per rail technology



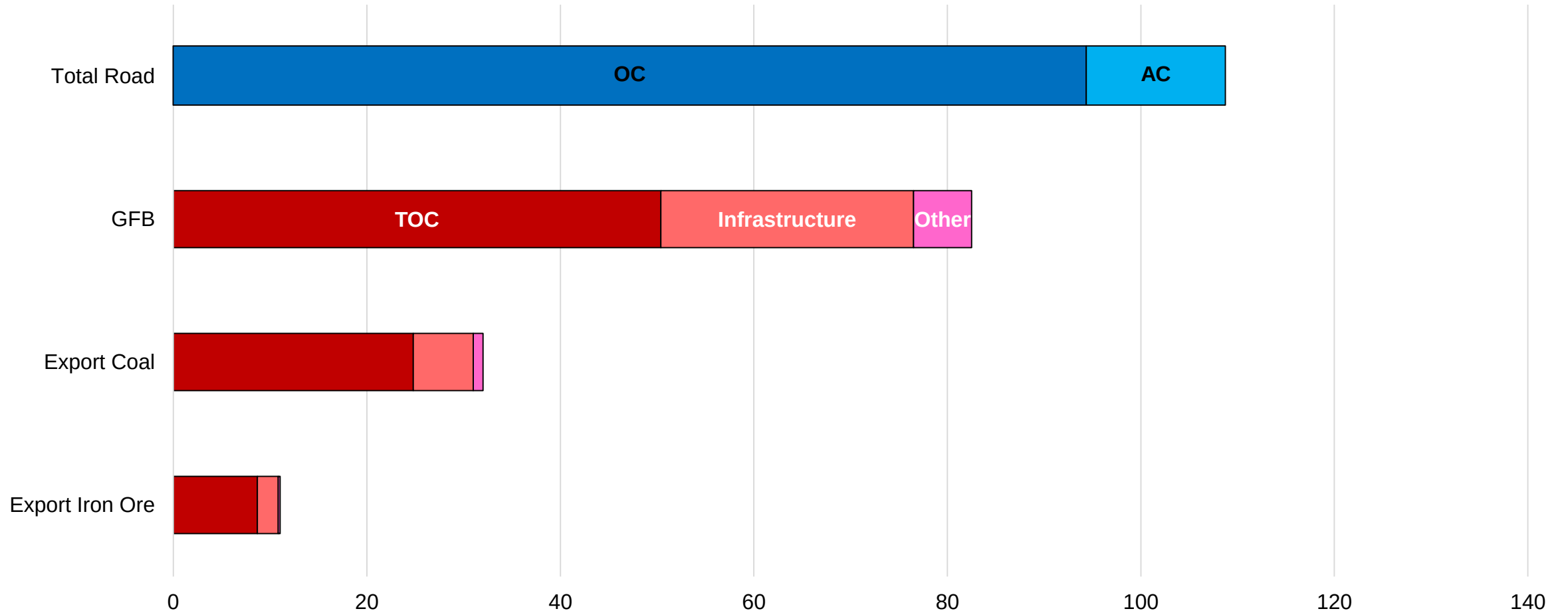
Source:

Rail cost components – High level model based on scientifically developed density curve, knowledge and experience;

Road cost components – Havenga, J.H. & Simpson, Z.P. 2018. Freight logistics' contribution to sustainability: Systemic measurement facilitates behavioural change.

Transportation Research Part D: Transport and Environment, 58:320-331. <https://doi.org/10.1016/j.trd.2016.08.035>

Road and rail cost components – per rail technology



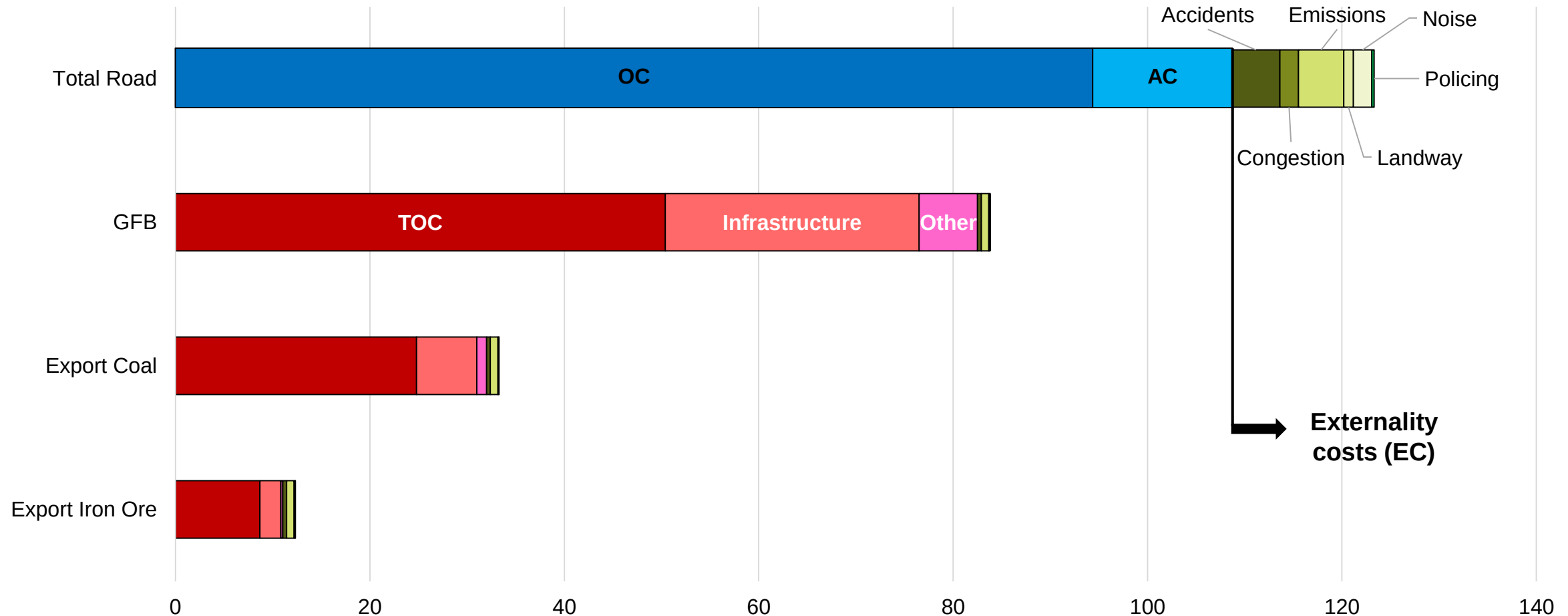
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Road and rail cost components – per rail technology



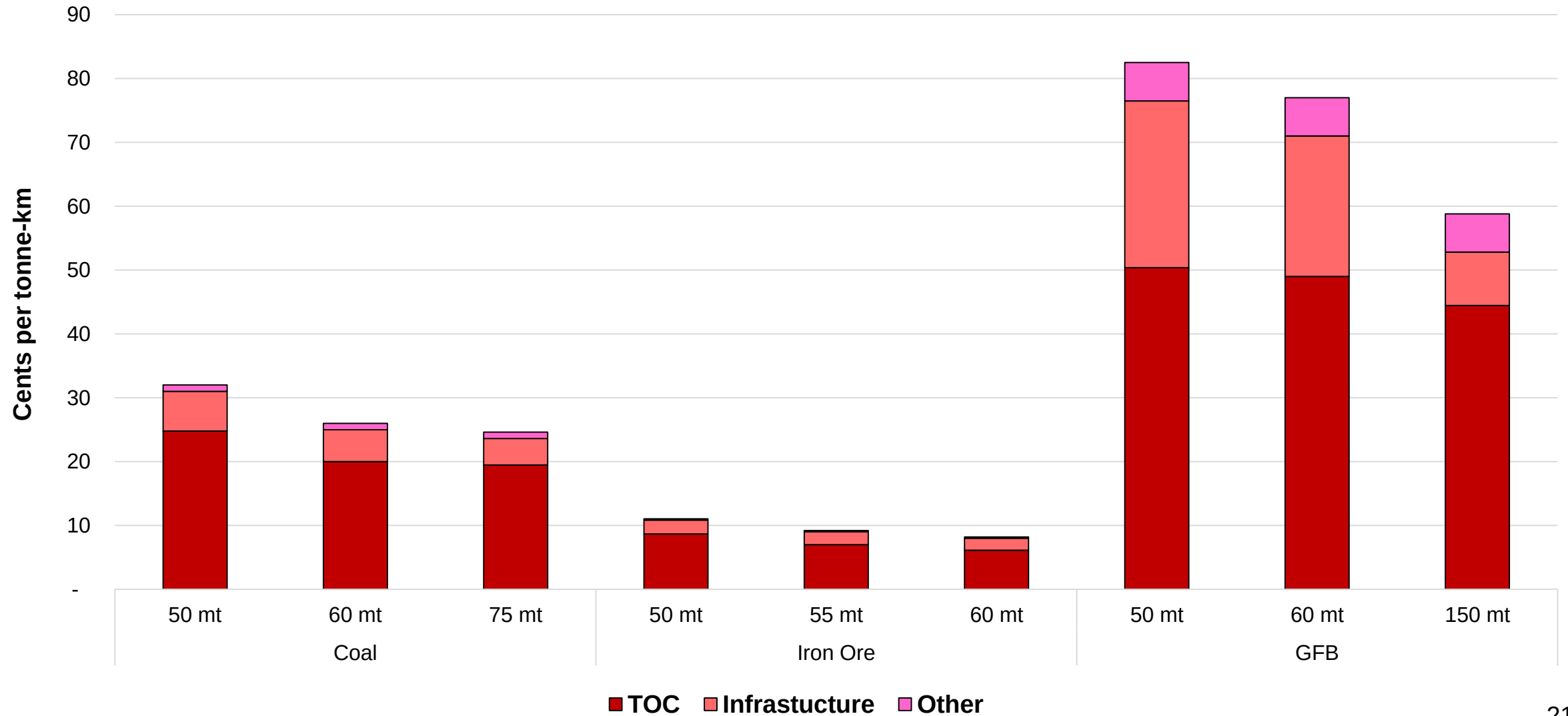
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Rail cost components – High level model based on scientifically developed density curve, knowledge and experience;

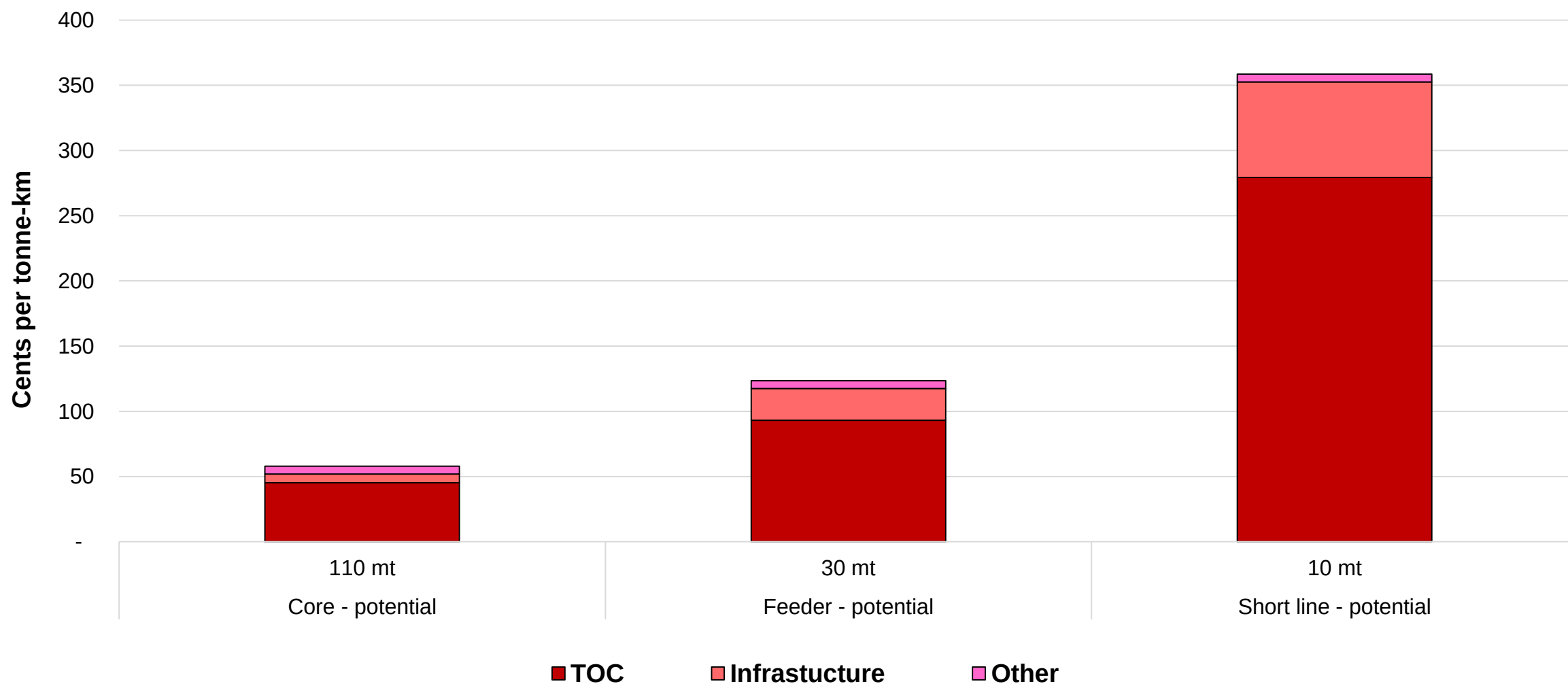
Road cost components – Havenga, J.H. & Simpson, Z.P. 2018. Freight logistics' contribution to sustainability: Systemic measurement facilitates behavioural change.

Transportation Research Part D: Transport and Environment, 58:320-331. <https://doi.org/10.1016/j.trd.2016.08.035>

And also by density within technology

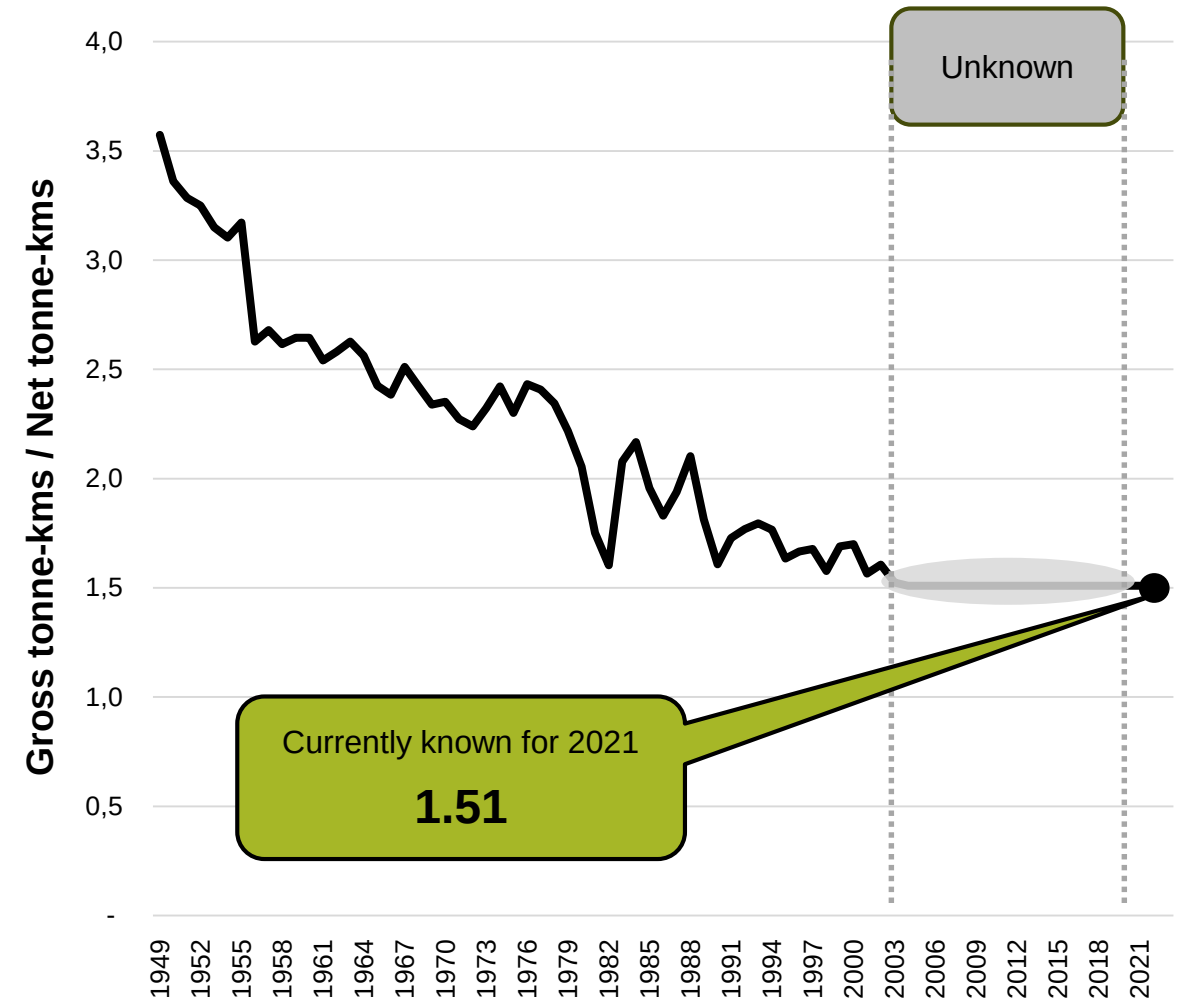
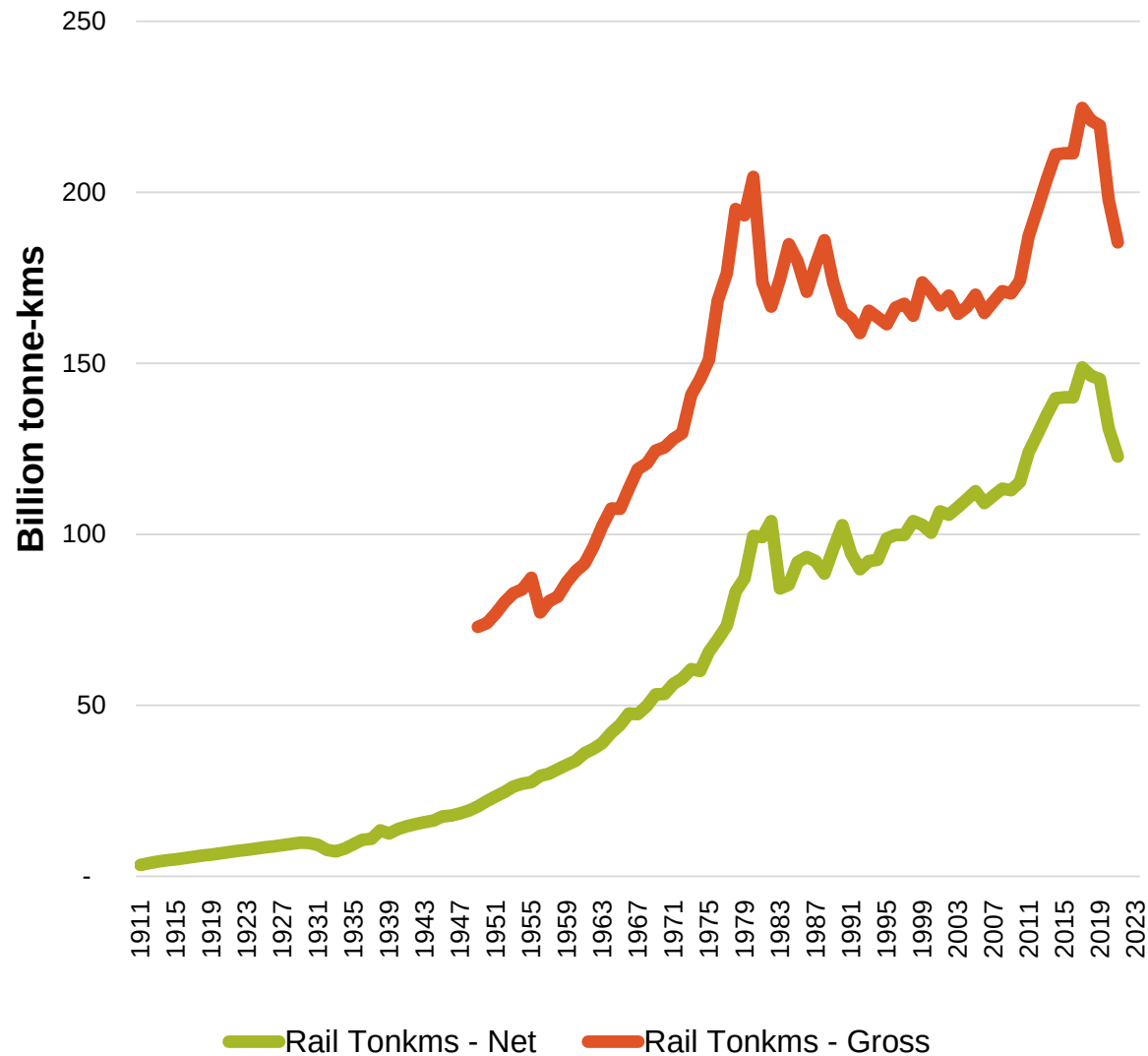


But rail cost is massively impacted by network tiers



But the actual cost structure will only be challenged at train operations/other level. What about Infrastructure? It could be as much as 50% to high.

Comparison between Gross and Net tonne-kms



Network statement response

- **The problem**
- **The promise**
- **The emergence of a rail to road strategy**
- **Why are we here? (Regulation history)**
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Specific rail regulation history



**RAIL ECONOMIC
REGULATOR**



**Revised Regulatory
Framework for the Rail
Economic Regulator**

3 June 2007

The role and development of South Africa's Freight Rail Sector

Prepared for Transnet Limited
by the Stellenbosch University
Centre for Supply Chain Management



December 2010

- Regulate service, price, price and quality
- Fair access, no monopoly
- Align with govt objectives

- Determine the rail role given government objectives
- Design an ideal logistics system
- Determine the role of rail in this
- Establish a single regulator (fair treatment of road and rail)
- Regulate based on a Masterplan

The recent history
is well known

But it is the the
tension between
the two viewpoints
that we
experienced

The DoT has
worked towards a
Masterplan based
ideal recently – the
process over the
last few years is
sound

It is now a
sequencing issue

Network statement response

- **The problem**
- **The promise**
- **The emergence of a rail to road strategy**
- **Why are we here? (Regulation history)**
- **The prize**
- **What now**

What we can gain

- Address one of our three largest production factor issues of our nation
- Increase:
 - Economic growth – R1 billion per day
 - Return to the Fiscus – R50 million per day
 - Jobs created – 500 per day

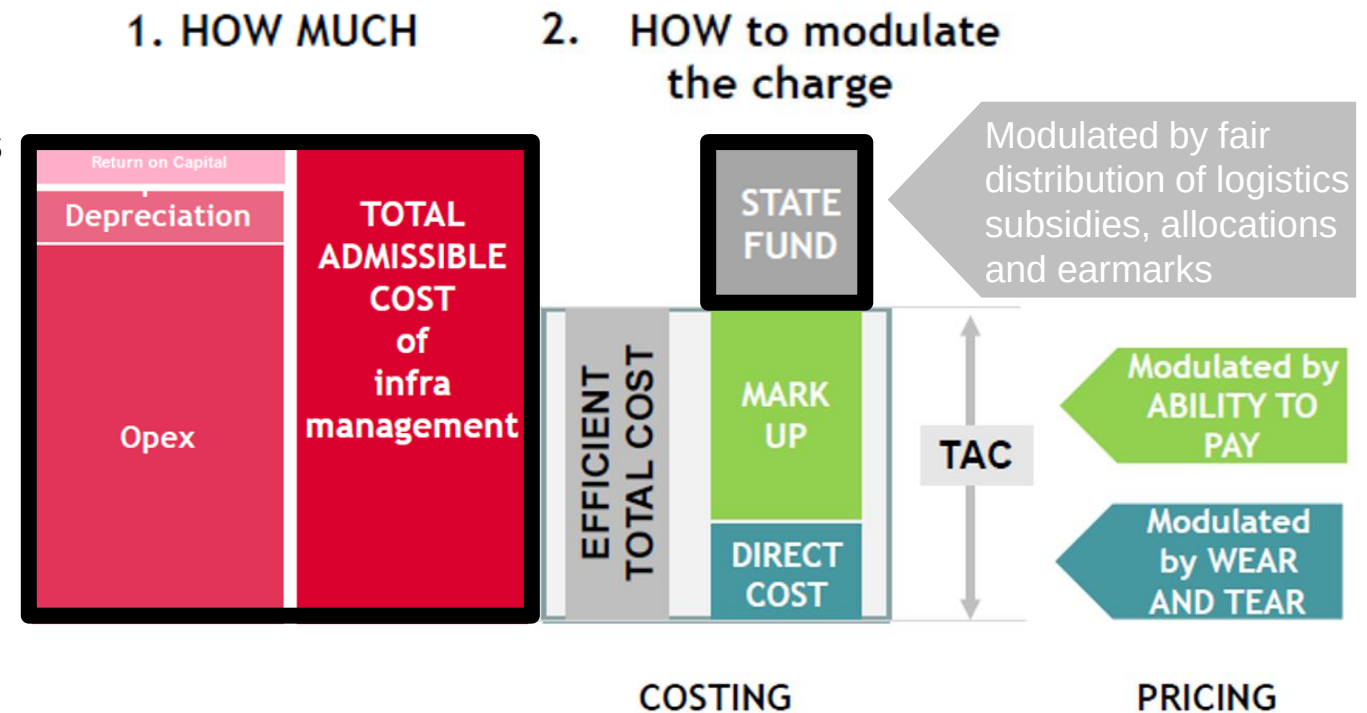
Save us from a failed state

Network statement response

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- **What now**

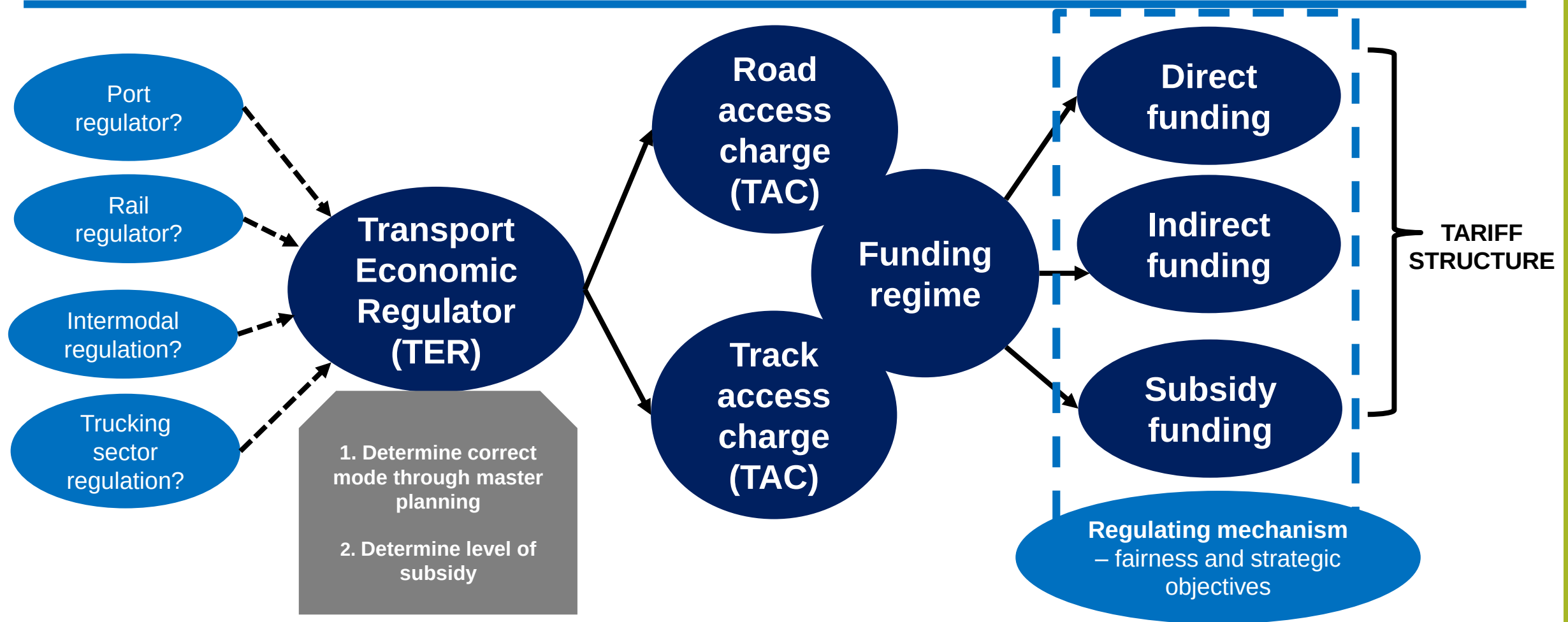
Response to TRIM Network Statement

1. We are solving for the nation/not Transnet
2. Actual cost to fix/operate/maintain network
3. Different parts of network separately
4. Cannot shock existing market - sharp price ups
5. We need a Transnet financial model
6. ROA should be out
7. If not value is – scrap + goodwill
8. Balance sheet shortfalls/loans – out
9. Externalities?
10. Formula that represent national objectives:
 - For road, rail and other logistics infrastructure



For all modes

Regulatory interventions



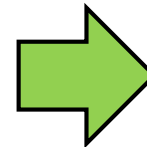
Subsidies should be fair, consider true cost benefit trade-offs and applied to the system.

Questions against a development strategy background

- Do we price for demand modulated by installed capacity or current performance?
- Do we price based on design parameters or vanilla?
- Do we want to pay for past mistakes?
- Do we want to extract returns?
- If so from roads/ rails or rail only?



- It will essentially not be possible to sort this out without:



National logistics vision

National logistics plan

Rail masterplan fit to this plan

Restructuring Transnet

Solving Transnet's debt problem

And a basic understanding of minimum four things. Different technologies, different densities, road to rail strategy including externalities, development state ideals. And then that current cost structure is untested.

Maybe only rent extraction should be regulated?

The future of rail has a direct link to mankind's survival

- A world without mass guided transport will be a failed world
- The same is true of South Africa

An active and vibrant railway system confers many benefits on society.

Olumyemi Osinbajo

Let us imagine ourselves transferred to our old friend, the railway carriage...

Albert Einstein

A failed railway will be a nail in the coffin of a failed state.



Thank you!

Table of costs

Rail segment	Scenario	TOC	Infrastructure	Other
Export Coal	50 mt	24.8	6.2	1.0
	60 mt	20.0	5.0	1.0
	75 mt	19.5	4.1	1.0
Export Iron Ore	50 mt	8.7	2.2	0.2
	55 mt	7.0	2.0	0.2
	60 mt	6.1	1.9	0.2
GFB	50 mt	50.4	26.1	6.0
	60 mt	49.0	22.0	6.0
	150 mt	44.4	8.4	6.0
Core potential	110 mt	45.3	6.7	6.0
Feeder potential	30 mt	93.1	24.4	6.0
Short line potential	10 mt	279.3	73.3	6.0

Source:

Rail cost components – High level model based on scientifically developed density curve, knowledge and experience;

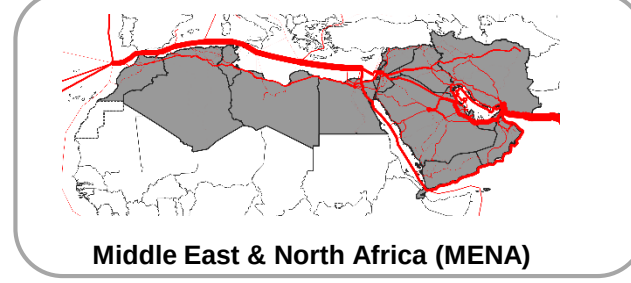
Who we are

How much of what moved where and how and at what cost?

- Tonnage
- TEUs
- Tonne-km
- Import/Export/Domestic
- Commodities
- Origins & Destinations
- Road/rail/pipeline/waterway
- Containers/bulk

Africa

Other

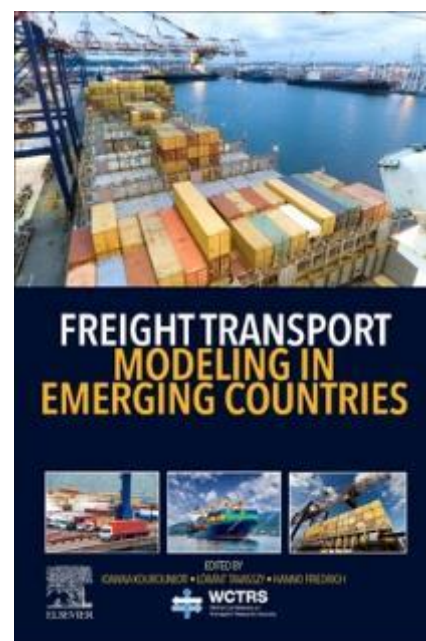
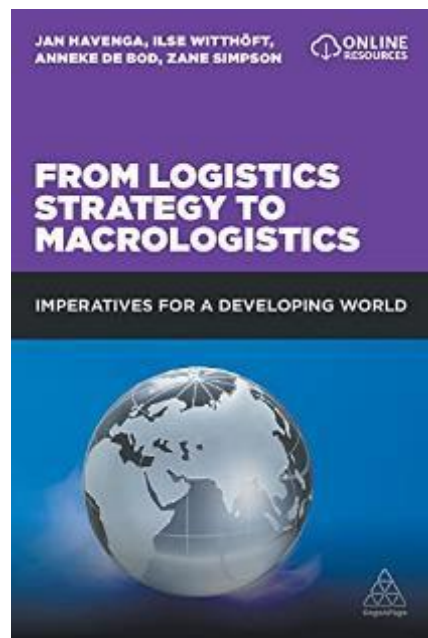


We include as many variables as possible:

- Costs
- Other activities: cash to cash cycle / warehousing / administration
- Including all externalities: emissions / noise / congestion / policing / accidents / land use



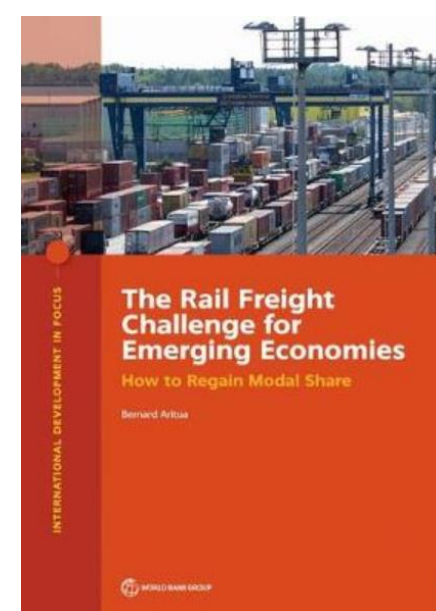
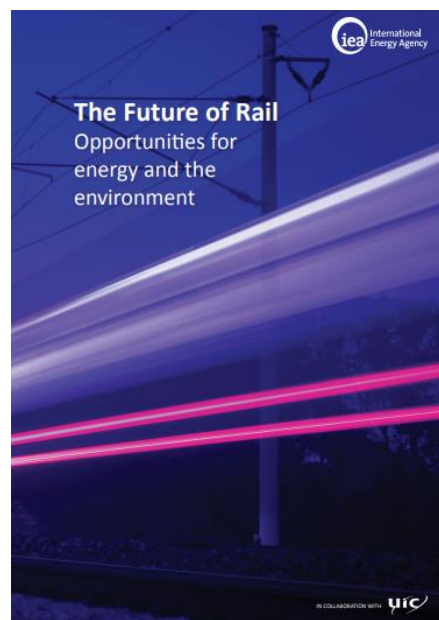
The objective is to map economies in motion and find the supply chain solutions for improvement.



37 peer reviewed articles
60 international peer review conference papers
20 invited/key note presentations on all continents
2 books, 2 chapters in peer reviewed books

Various reports
(latest 1st road to rail strategy for RSA and Transnet's effect on economy)

Direct recommendations from the best in this field.
Including "father" of Freight Flow modeling as well as the co-creator of the WB LPI



Deconstructed rail network view – GAP?



Bulk mineral corridors (2 241 km)

Heavy haul “conveyor belts” to bulk ports
Limited intermodal competition
Requires high density and homogeneity

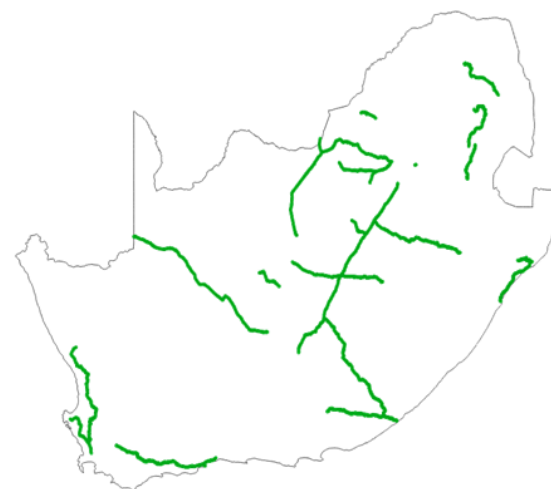
R30bn → \$1.5-2bn



Core rail network (4 818 km)

Major economic connections (ports and metros)
Mostly high-density or strategic connection role
Diverse commodities and segments
Critical in road-to-rail effort
Important role in supply chains

R80bn → \$4-5bn



Feeder network (5 316 km)

Functionally and technological different
Important support for the core
Strategic role
Can sometimes replace the core in emergencies

R60bn → R\$3-4bn



Short lines (6 532 km)

Low volume potential
Could have important social or industry value
Minimum approach is ROW protection
Fair and equitable dispensation should be sought

R20 bn \$1bn

The maintenance shortfall might be approaching R200 bn or \$10 billion.

Infrastructure interventions

